

Planning for Cruise Ship Resilience: An Approach to Managing Cruise Ship Impacts in Haines, Alaska

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Resilience theory has management implications useful when planning for cruise ship tourism. In small coastal communities, cruise ships often provide welcomed economic incentives that can bolster a waning economy. However, in some coastal communities, the magnitude and intensity of passenger visits can reduce social resilience and induce an economic regime shift that leads to rapid socioeconomic reorganization. The implications of such a regime shift are a loss of economic diversity, reduced social resilience, and a loss of social capital. Planning for cruise ship resilience increases the likelihood of successful coping strategies and addresses the socioecological changes inherent to a cruise ship destination. Both Holling's adaptive cycle and the four principals for building resilience established by Berkes and Seixas are fundamental to a successful management plan.

Keywords Alaska, community resilience, cruise ship tourism, social resilience, tourism planning

When the fire is contained and managed within the hearth it offers beauty and comfort to the household. When it runs out of control it can destroy the very household it was designed to support. (Murphy, 1986)

Resilience, Community, and Cruise Tourism

Resilience Theory

Resilience theory provides a lens for evaluating the overall function of socioecological systems undergoing change (Folke, 2006). The resilience of a system refers to the relationship between that system's components and the capacity of those relationships to endure change and maintain overall function without ceasing to exist (Walker et al., 2004). The concept of resilience is closely linked with the principals of sustainable development. Following Holling (2001) and Ruiz-Ballesteros (2010), "sustainable" refers to the capacity to build adaptive competencies, while "development" is the production of opportunities. Resilience thus provides a community with the ability to both survive and maintain the capacity to benefit from opportunities (Ruiz-Ballesteros, 2010). The opposite of resilience is vulnerability (Folke, 2006). Vulnerability results when the magnitude or intensity of change to a

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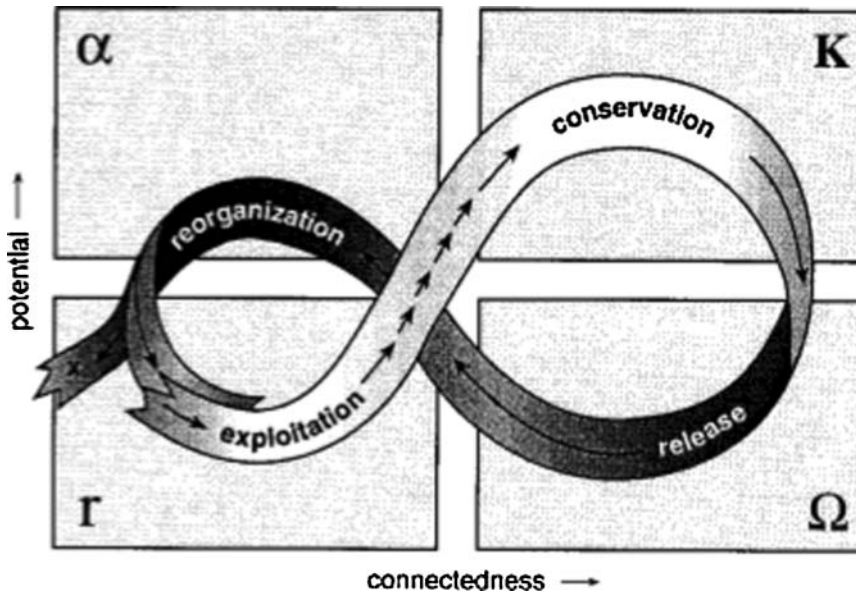


Figure 1. The adaptive cycle of a socioecological system and the four system states (r , K , Ω , α). Fixed arrows indicate a rapid rate of change (Gunderson & Holling, 2002). The Y-axis indicates the potential for change based on available resources and the X-axis indicated the degree to which the system is interconnected. High potential and a large degree of interconnection is indicative of a vulnerable system with low resilience (Gunderson & Holling, 2002). From *Panarchy* edited by Lance H. Gunderson and C. S. Holling. Copyright © 2002 Island Press. Reproduced with permission of Island Press, Washington, DC.

system is beyond the adaptive capacity of that system's components (Ruiz-Ballestros, 2010 citing Folke, 2003, 227). Holling (2001) suggests a complex system such as a community undergoes an adaptive cycle composed of three fundamental properties that shape that system's future (Figure 1).

- **The wealth of a system** is the potential for change based on the range of future options.
- **The controllability of the system** refers to the connectedness of the system's components and the ability to change.
- **The adaptive capacity of the system** is the resilience and ability to withstand perturbations and maintain the same internal functions while undergoing change.

Consider Figure 1 in the context of community development. During the slow exploitation phase (r), social capital is built. Community connections and trust are established, and social and economic diversity build resilience. As the system flows toward the conservation phase (r - K), resources become less available (Holling, 2001). What once were sources of economic diversity become highly interconnected thereby reducing resilience and increasing vulnerability (Holling, 2001). The rapid flow from the conservation phase toward release (K - Ω) is triggered by a disturbance such as an economic downturn or social upheaval and results in system collapse. The system then undergoes rapid reorganization (Ω - α) and the community grapples for stability and attempts to rebuild. According to Holling (2001), during the reorganization phase, new combinations for community structure can be established and the opportunities for experimentation with new social combinations

exist. The repetitions of the adaptive cycle increase the predictability of the outcome of reorganization. The exit loop (X) signifies a departure from reorganization and the adaptive cycle and digression into a less organized state (Holling, 2001).

Although resilience theory mostly concerns understanding how a system responds and adapts to change, there is also a management component to the discipline (Folke, 2006; Ruiz-Ballestros, 2010). Considering community resilience when planning for cruise ship tourism can lead to effective and sustainable pathways for coping with future surprises and changes inherent to humanity (Folke, 2006). In the case of Haines, Alaska, planning for community resilience may prolong exploitation and mitigate the increased vulnerability associated with rapid development.

Community

A community is broadly defined by Ruiz-Ballestros (2010) as a socio-anthropological framework for organization and meaning that contains political institutions and governance relationships. As well, in a community myriad and diverse social connections, networks, and practices exist. Change is inherent to communities whether the source is natural or anthropogenic. In the face of change, a resilient community will find opportunities for development or diversification. On the other hand, a vulnerable community lacking adaptive capacity may be devastated or cease to function when presented with the same degree of change (Folke, 2006; Ruiz-Ballestros, 2010). Adger (2000) defined social resilience as the ability of human communities to cope with stresses and disturbances to their social infrastructure while maintaining the integrity of community function despite environmental, social, economic, and political disruption. The social resilience and adaptive capacity of rural or isolated communities is of particular management concern given the vulnerability of these systems (Ruiz-Ballestros, 2010). In growing numbers, tourism ventures are interested in isolated regions (Ringer, 2006). In Alaska for example, rural towns once dependent on a patchwork economy of timber harvesting, fishing, and mining are now abandoning these traditional economic sources in place of tourism (Cervený, 2004; McDowell Group, 2002).

Cruise Tourism

Alaska is easily accessible to mass tourism via the cruise ship industry and provides a gateway to what many visitors consider to be “the final frontier” of untouched American wilderness. As such, the tourism appeal is the opportunity to view free-ranging wildlife, the pristine beauty of wilderness areas, and experience the remote existences and colorful history of communities removed from the cultural epicenters of developed regions. As the cruise industry grows annually in Alaska, planning for cruise ship resilience is of increasing importance to the maintenance and function of the small coastal communities along the cruise track.

Cruise ships bring economic gains but also a suite of management issues (Garin, 2005; Zapata-Aguirre & Brida, 2008). Between 1990 and 2000 cruise tourism in Haines grew at a rate that outpaced infrastructure development and stressed community function and social connections (Cervený, 2004; McDowell Group, 2002). The massive influx of passengers caused excessive crowding and congestion in the town center, caused changes in revenue sources and distribution throughout the local economy, and threatened the ecology of the town’s wilderness attractions (McDowell Group, 2002). Tourism in Haines in the 1990s exemplifies the need for a management plan that considers cruise ship resilience.

Berkes and Seixas (2005) present four criteria necessary for building resilience in socioecological systems: (1) learning to live with change and uncertainty; (2) nurturing

diversity for reorganization and renewal; (3) combining different kinds of knowledge; and (4) creating opportunity for self-organization. This framework, although originally applied to lagoon and estuarine systems, provides a valuable method for evaluating and building social resilience in small coastal communities facing the pressures of cruise tourism (Ruiz-Ballestros, 2010). With this intent, this article seeks to apply Berkes and Seixas resilience-building framework to the Haines tourism management plan developed by the McDowell Group in 2002. The addition of resilience theory to the existing management recommendations provides a more comprehensive approach for coping with cruise tourism in small coastal communities.

The Resilience History of Haines, Alaska

Haines, Alaska, by definition is a coastal community “fundamentally reliant on coastal natural capital in one form or another” (Gibbs, 2008). Haines is located 85 miles north of Juneau in the upper portion of Lynn Canal in Southeast Alaska (Figure 2). As of 2000,

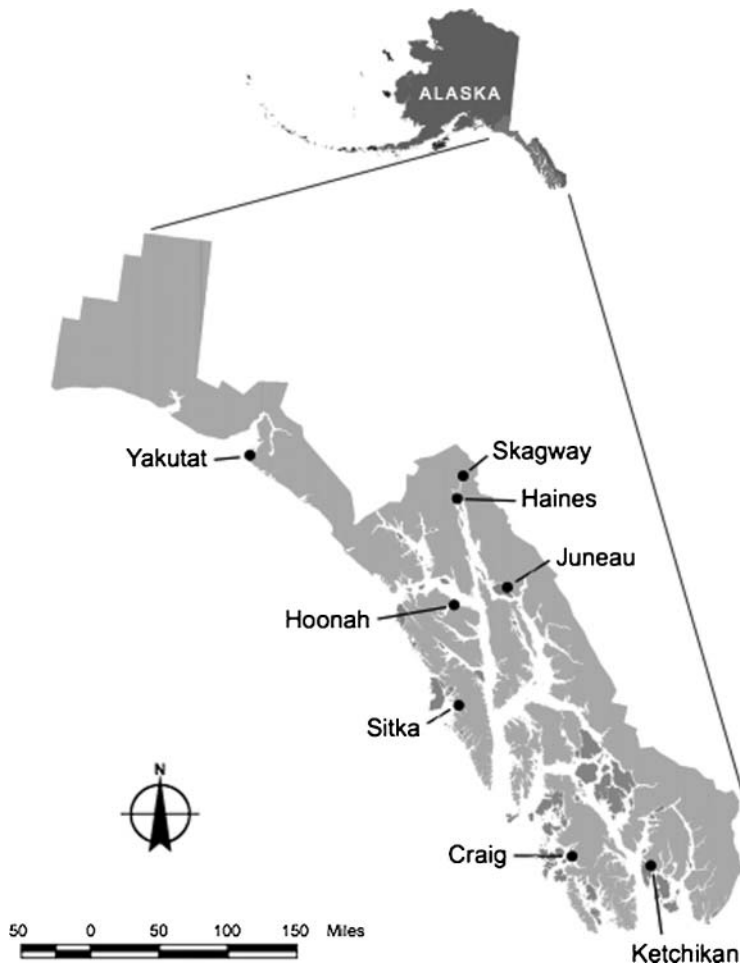


Figure 2. The major ports affected by cruise tourism in Southeast Alaska (Cervený, 2004).

the population of Haines was 2,392 people (U.S. Census Bureau, 2009). Historically, Haines was a working town and economically bound to timber, fishing, and mining (Haines Chamber of Commerce, 2009; McDowell Group, 2002). Since the beginning of the 20th century, the economy was fairly well diversified and composed of both regional industry and outside investors (Cervený, 2004). Like Craig, Hoonah, and Skagway—three similar Southeast Alaskan towns—Haines had its share of economic difficulties. The cyclical trends in the region's boom-and-bust economy due to fluctuations in resource availability made economic depression commonplace (Cervený, 2004).

Industry in Haines came and went. Beginning in the late 1800s, the foundations of the town's economy were built upon commercial fishing, canneries, military, and transportation projects, as well as the mining industry established during the gold rush (McDowell Group, 2000). Many of these industries lasted well into the 20th century and provided work for residents of Haines and those drawn to Alaska from the lower 48 states by the booming fisheries and industrial opportunities. Fort William Henry Seward, established at the turn of the century, provided both jobs and community functions until its deactivation in 1946. The jobs lost in the military closure were supplemented by jobs created by construction on the Haines Highway and the Haines-to-Fairbanks military jet fuel pipeline. The highway served as a critical thoroughfare between Haines and points to the north such as Fairbanks and the Yukon. Ultimately, this highway provided access for tourists to visit Haines (McDowell Group, 2002). Also during this period, the timber industry grew to be a substantial source of economic well-being for the region. As a result of swings in natural resource availabilities and the boom–bust economic cycles of the “patchwork” economy, residents of Haines grew familiar with economic instability (Cervený, 2004). In this period of history, prior to large-scale cruise tourism, the social and economic structures in Haines embodied the early stages of Holling's exploitation phase of the adaptive cycle: resilient to loss of function despite economic downturns while maintaining adaptive capacity.

The Rise of Tourism (r-K)

In the early 1980s, declines in fish stocks and the closure of a key timber mill led to economic recession (McDowell Group, 2002). The resulting economic downturn coupled with the completion of the Haines-to-Fairbanks highway caused interest in tourism to grow in Haines (Cervený, 2004). In the 1970s, continued improvements to the highway, the expansion of ferry routes to include Haines, as well as regular visits by small cruise ships made tourism a viable economic supplement in the region. In 1983 the community acknowledged tourism as a viable source of economic revival and pushed for a larger role in the market (Cervený, 2004). Quickly, tourism grew into a valuable part of the waning economy in Haines. Small cruise ships as well as larger steamships and ferries regularly visited the municipal dock (Cervený, 2004). Royal Caribbean and Exploration Cruises lightered passengers ashore several times per week to walk the muddy town streets in search of souvenirs and an Alaskan experience. The investment in tourism resulted in 20,000–30,000 visitors annually and the revival of the small town's economy (Cervený, 2004). People moved to Haines, tourism-based companies appeared, and businesses expanded to accommodate the new influx of visitors (McDowell Group, 2002). In 1988, Exploration Cruises went bankrupt and the number of annual cruise ship visits to Haines dropped from 100 to 20 (Cervený, 2004). This economic loss was compounded in 1990 when a second timber mill closed. Tourism, above other economic sources, was again looked to as a solution to recession (Cervený, 2004).

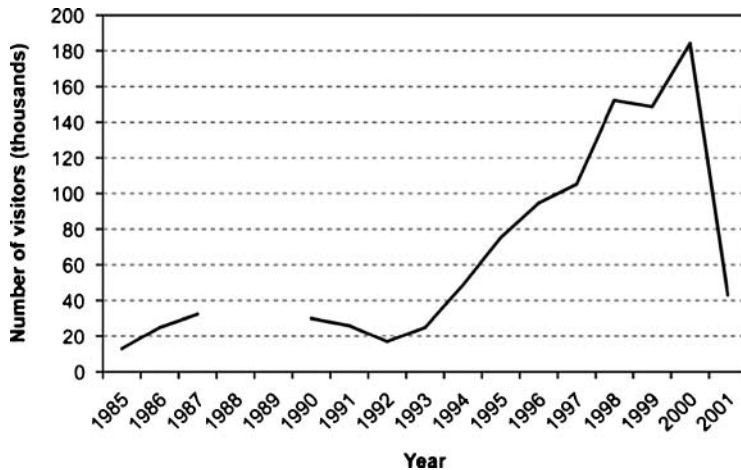


Figure 3. Cruise ship passenger visits to Haines 1985–2001 (Cerveny, 2004).

In 1994, Haines formally committed to the cruise industry and the municipal dock was renovated to accommodate larger ships (McDowell Group, 2002). Concurrent with the expansion of infrastructure in Haines, the cruise ship industry as a whole grew. Between 1995 and 2000, the number of ships in service increased by 15% annually and Alaska became a lucrative and popular cruising destination on par with the Caribbean (Cerveny, 2005; Ringer, 2006). In 1995, the completion of the Port Chilkoot cruise ship dock in Haines marked a turning point in the small town's relationship with cruise tourism. The subsequent growth in annual visitors to Haines was dramatic (Figure 3). In 1994, while the dock was still under construction, 48,000 people visited annually. In 1995, ships owned by Norwegian Cruise Lines and Royal Caribbean carried 75,000 tourists to the small town. In 1996, 95,000 people arrived. In 2000 the number of cruise ship passengers visiting Haines was estimated to be over 180,000 people (McDowell Group, 2002). The mass-influx of tourism dollars overshadowed economic contributions from the fishing and timber industries and the town's focus shifted to retail, tour operations, and service (Cerveny, 2004; McDowell Group, 2002).

The Impacts of Cruise Ship Tourism (K-Ω)

The arrival of large cruise ships to this Alaskan town in the late 1980s brought substantial economic benefits at a time when primary sources were in decline (McDowell Group, 2002). Between 1989 and 1999, the number of visitors travelling to Alaska by ship increased eight times, from 20,000 to 159,000 people annually (Figure 3) (McDowell Group, 2000). During this timeframe however, the number of jobs in Haines increased by only 5%, with zero net growth between 1997–2000 (McDowell Group, 2002). Despite the lack of new job creation, the service industry boomed. The amusement and recreation services sector increased 73% and retail increased 15% (McDowell Group, 2000). In 1990, 11% of Haines residents were employed in the tourism industry. By 2000, that number more than doubled to 26%, or one in four residents (Cerveny, 2004). The slow job growth and stagnation across all traditional economic sectors despite the rapid tourism expansion is indicative of a large-scale shift in economic focus from traditional economic sources to tourism and the service industry. The community in Haines grew reliant on cruise tourism, and many of the traditional sources of

income were abandoned for tourism-based jobs (Cervený, 2004; McDowell Group, 2000). Although the town expressed interest in continued expansion of the cruise industry, planners recognized a need for conservation and considered ways to slow the growth of tourism in Haines (McDowell Group, 2002).

The impacts of increased tourism were both social and environmental (Alaska Coastal Management Program, 2006). The accumulation of garbage, improper disposal of sewage and congestion in the city center due to both foot traffic and RV tourism strained infrastructure. Resource competition developed between residents and tourists and heightened tensions. The environmental impacts of tourism were also numerous. Tourism caused increases in pollution from both car and vessel exhaust, concerns over the discharge of black and gray water, and placed pressure on the natural destinations in Haines. Moreover, residents were concerned the ecological pressures of tourism would alter the migratory habits of salmon (Alaska Coastal Management Program, 2006).

In response to these impacts, town planners proposed a cruise ship visitation cap designed to limit the number of vessels permitted to call on Haines. In addition to the cap, the committee proposed a 4% tax on tours and charters and a 4% bed tax. This strategy predicted an additional \$201,733 in annual revenues from cruise traffic (Cervený, 2004). Residents hotly debated this initiative despite the obvious financial incentives and the need for a mechanism for tourism regulation (Alaska Coastal Management Program, 2006). Supporters saw the taxation system as a means of providing greater economic benefits to the community and as a funding source for necessary infrastructure improvements. The cruise ship cap was seen as a definitive method for regulating passenger numbers. Moreover, a tax on tourism was in line with taxation on other preexisting industries in Haines (Cervený, 2004). Those opposed feared any tax on tourism would restrict economic development, disrupt the tenuous relationship the town had with the cruise industry and make Haines a less appealing destination (Cervený, 2004). When put to a vote in 2000, the initiatives passed by a narrow margin.

Almost immediately upon implementation, the number of cruise ship visits to Haines declined (McDowell Group, 2002). One small cruise company removed Haines from their itinerary citing the presence of many large cruise ships and congestion as their reason for leaving (Cervený, 2004). Shortly thereafter, Norwegian Cruise Lines and Princess Cruise Lines terminated service to Haines, citing an unfriendly attitude of residents and the prospect of more lucrative and economically beneficial visits elsewhere. In 2001, at the height of the town's tourism industry, Royal Caribbean, the largest contributor to the town's tourism economy, discontinued service to Haines (McDowell Group, 2002). Royal Caribbean claimed their decision was made in response to rising fuel costs and more lucrative options in Skagway (Cervený, 2004). By removing Haines from their itinerary, Royal Caribbean estimated 1.45 million dollars in fuel savings (Cervený, 2004). However, despite the reasons presented by the cruise companies, many residents cited the tourism tax as the cause (Cervený, 2004). Furthermore, some perceived Royal Caribbean's decision a tactic to show other Alaskan towns a display of corporate might (Cervený, 2004).

The Royal Caribbean decision caused the number of annual passengers to drop from 180,000 to 43,000 and the local economy was devastated by a disturbance of this magnitude (Figure 3) (Cervený, 2004; McDowell Group, 2002). In the context of Holling's adaptive cycle, the single-sector economic dependence developed during the exploitation phase between 1990–2000, increased the community's vulnerability and reduced adaptability (Beatley, 2009; Holling, 2001). The rapid decline in cruise passengers was adequate to drive the transition from conservation into the release phase (K- Ω). Rapid social reorganization in Haines followed. As noted by Holling (2001) reorganization after a disturbance presents a

community with the opportunity to build new social structures, processes, and innovations. A resilient community is able to continuously adjust and utilize disturbance to develop adaptive capacity (Folke, 2006). Although Haines was accustomed to economic downturns, Royal Caribbean's departure provided a new opportunity to establish management strategies to cope with cruise tourism. Importantly, the Royal Caribbean decision taught residents of Haines that cruise ships are a fickle economic source with the potential for large social and economic impacts worthy of a tourism management plan.

Planning for Cruise Ship Resilience

In 2002, one year after the Royal Caribbean decision, The McDowell Group, a research and consulting firm in Southeast Alaska, aided reorganization in Haines. The firm was hired to develop a management plan to guide tourism development and improve the local economy. Despite the unpredictability of cruise ship dependence, the majority of surveyed residents wanted to either continue cruise tourism development or cap ship visits at the 2000 level despite the potential for economic downturn (McDowell Group, 2002). Haines required conservative and planned development of their tourism industry. The plan drafted by the McDowell group recommended the following management objectives based on community surveys and analysis of the tourism industry at the time:

- The community prefers a moderate long-term economic growth rate.
- Economic benefits of employment, personal income, and business income shall be directed primarily at current local residents and businesses.
- The community prefers a tourism economy that relies on several markets rather than a single dominant one.
- The community prefers development that has wide distribution of economic benefits.
- The community prefers a strategy that will stimulate the economy in the near term while building a more solid economy in the long term.
- Tourism management includes proactive development designed to enhance, accommodate, and mitigate the various impacts of tourism.

Although these recommendations provide sound economic planning advice, the importance of building and maintaining social resilience is largely overlooked. The work conducted by Berkes and Seixas (2005) and Ruiz-Ballestros (2010) on socioecological systems of lagoons indicates the importance of the development and maintenance of social resilience as a valuable management component. Therefore a successful tourism management plan must address economic issues but also draw on principals of sustainable development and resilience.

Furthering the work of the McDowell Group, tourism management in Haines can also be considered in the context of Berkes and Seixas (2005) four criteria for building social resilience. A resilience perspective on tourism management may provide social learning and management implications that increase the likelihood of this plan achieving success (Ruiz-Ballestros, 2010). Management success in this community should be measured in both economic fortitude and in community resilience and adaptive capacity.

Learning to Live with Change and Uncertainty

Uncertainty and change are fundamental components of humanity and tourism is a potent progenitor of these components. Prior to large-scale tourism, Haines was a resilient community capable of adapting to swings in the economy. The economic history was defined

by a patchwork of resource and tourism based sources in constant flux. Holling's adaptive pathway provides a graphic representation of how the shift to mass tourism increased community vulnerability. Ultimately, the economic disturbance caused by Royal Caribbean spurred community reorganization.

Learning from a catastrophe is essential to building resilience (Ruiz-Ballestros, 2010). Of equal importance is the ability to apply those lessons effectively to mitigate future devastation (Berkes & Seixas, 2005). The expansion of cruise tourism in Haines was unchecked and the social and economic impacts associated with large-scale cruise tourism were unknown at the time. Furthermore, management attempts were made too late in the industry's evolution (McDowell Group, 2002). Appropriately, The McDowell Group's tourism management plan calls for the steady development of the tourism industry, diverse markets, and a broad distribution of economic benefits. Unmentioned is the development of rapid feedback mechanisms and strong leadership capable of guiding reorganization and adaptation (Ruiz-Ballestros, 2010). It is clear cruise tourism has the power to alter the function and social dynamics within a community (Ringer, 2006). Therefore a successful shift toward tourism also hinges on stakeholder buy-in and the potential for unified decision-making (Miller & Auyong, 1991). In Haines, prior to the loss of Royal Caribbean, relations between tourism brokers and local residents were strained over differing beliefs about how to manage the industry (Cervený, 2004). This lack of consensus is indicative of reactionary planning, poor feedback mechanisms, and failure to anticipate the effects of tourism (Folke, 2006).

Nurturing Diversity for Reorganization and Renewal

An intact social memory of past disturbances is critical to forming a logical response to change in the present (Berkes & Seixas, 2005). Social memory is a socioecological system's ability to recall effective responses to passed disturbances for use in management (Ruiz-Ballestros, 2010). Residents of Haines recognize the community continues to undergo change and the importance of planning for a future (McDowell Group, 2002). Furthermore, the community is aware of the pitfalls of adopting a reactive management strategy to cruise tourism (McDowell Group, 2002). Thus by nurturing this understanding and promoting a diversity of social and economic networks a community can develop a robust adaptive capacity. Additionally, tourism success requires developing methods for coping with change and disturbance, clear communication pathways across scales, and ways for resolving disputes (Ruiz-Ballestros, 2010).

Combining Different Kinds of Knowledge

Both the use of local knowledge and outside resources can help a community build and maintain social resilience (Berkes & Seixas, 2005). As the cruise industry expands worldwide and tourism research progresses, myriad examples of management successes and failures exist. The use of outside resources will provide Haines with a suite of cause and effect examples of how tourism affects communities. Planning for resilience should also be a participatory process that involves all affected stakeholders. Decisions made on a community level and across scales should be transparent and should build consensus, trust and foster creativity (Berkes & Seixas, 2005). Moreover, stakeholder involvement promotes a diversity of perspectives that combine local and scientific knowledge to arrive at the best management decision (Ruiz-Ballestros, 2010).

Creating Opportunity for Self-Organization

Addressing disturbance on a community level requires above all else, social capital (Berkes & Seixas, 2005). Social capital is a measure of the cohesiveness of a community and the ability to collectively cope with a disturbance through “trust, well-developed social networks, and leadership” (Beatley, 2009; Walker & Salt, 2006). To build social capital in Haines the community must have methods for self-organization (Berkes & Seixas, 2005). In other words, in the absence of leadership, Haines must develop the capacity to form a logical and coordinated response to stressors that transcends social barriers and functions across scales. Specifically, a self-organizing community should have the capacity to address external stresses, manage conflict, and fairly allocate resources (Berkes & Seixas, 2005). Self-organization, however, is fundamentally dependant on strong nurturing strong leadership across scales and establishing collective goals for management.

Conclusion

Coastal communities are inherently vulnerable. The impacts of cruise tourism present a double-edged sword in which economic gains must be balanced with the risk of losing economic freedom, cultural autonomy, and social resilience. As illustrated in Haines, Alaska, cruise tourism brings substantial economic benefits to a region in which industry was historically unstable and economically depressed. However, it is also clear the need for planners to consider the impact these large vessels and the tourists they bring will have on the region when accepting cruise tourism as a means of economic revival or as a supplement to an existing economy. The magnitude and intensity of this form of tourism is potent and capable of dramatically altering social and community dynamics (Ringer, 2006). By presenting the resilience history of Haines, Alaska in terms of Holling’s adaptive cycle, this article hopes to highlight the importance of planning for cruise ship resilience in coastal communities and to present a framework for thinking about how resilience theory can form an integral component in a tourism management plan. Furthermore, the events in Haines and the work of the McDowell group make a strong case for proactive management of cruise tourism. As shown in Haines, reactive management measures are commonly ineffective and often enacted too late to mitigate impacts. Via proactive management, attention to the principals of sustainable development and the maintenance of social resilience, coastal communities can exploit the benefits of a cruise ship economy while enduring the associated impacts.

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